

Tri-Level Hybrid DDQN-ALNS: A Hierarchical Learning-Augmented Metaheuristic for the Vehicle Routing Problem with Time Windows

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ABSTRACT The Vehicle Routing Problem with Time Windows (VRPTW) is a fundamental combinatorial optimization problem in logistics, characterized by a hierarchical objective that prioritizes minimizing the total number of deployed vehicles (N_V) before minimizing total travel distance (D_T). While classical Adaptive Large Neighborhood Search (ALNS) remains a widely used and competitive metaheuristic paradigm for vehicle routing, baseline implementations relying on stateless heuristic score aggregation and static acceptance criteria can encounter search stagnation in complex spatial-temporal topologies. Conversely, pure end-to-end Deep Reinforcement Learning (DRL) methods face feasibility challenges under hard time windows and out-of-distribution generalization barriers. To bridge these paradigms, this paper presents Tri-Level Hybrid DDQN-ALNS, a learning-augmented optimization framework structured into three coordinated tiers: (i) a Macro Decision Tier (Plateau Controller) formulated as an Approximate Semi-Markov Decision Process over an Aggregated State Space (Aggregated SMDP) to monitor search stagnation and govern regime transitions, (ii) a Micro Selection Tier where a Dueling Double Deep Q-Network (DDQN) with an explicit normalized Softmax-Entropy policy-concentration gate learns state-conditioned destroy-repair pairings, complemented by a Learned Acceptance Criterion (LAC) that operates as an online delayed-outcome predictive acceptance override over an $H_{lac} = 80$ lookahead horizon without vetoing valid Simulated Annealing acceptances, and (iii) a Heuristic Execution Muscle leveraging $O(1)$ incremental forward-time-slack feasibility queries per candidate insertion, a Contrastive Graph Neural Network (GNN) spatial edge predictor to prune unpromising insertion arcs, polynomial Generalized Ejection Chains (GEC), and periodic time-limited Set Partitioning recombination over elite routes via HiGHS with certified lexicographic penalty bounds. We evaluate the proposed framework across 74 standard Solomon-100, Homberger-200, and Homberger-400 benchmark instances under strictly isolated cold-start execution protocols. Tri-Level Hybrid DDQN-ALNS achieves statistically significant vehicle fleet reductions (31 Wins / 40 Ties / 3 Losses, Wilcoxon $p = 6.91 \times 10^{-7}$); on the 54.1% fleet-matched subset ($N_{matched} = 40/74$), it secures an average -3.73% travel distance reduction ($p = 9.31 \times 10^{-8}$) while strictly preserving 100% operational constraint feasibility.

INDEX TERMS Adaptive Large Neighborhood Search, Deep Reinforcement Learning, Double DQN, Semi-Markov Decision Process, Metaheuristics, Vehicle Routing Problem with Time Windows.

I. INTRODUCTION

The Vehicle Routing Problem with Time Windows (VRPTW) is a foundational NP-hard combinatorial optimization

tion problem underpinning modern freight logistics, last-mile parcel delivery, and sustainable distribution networks [1], [2]. In addition to the combinatorial explosion inherent in classical routing, VRPTW enforces hard customer time-window constraints: each vehicle route must simultaneously respect vehicle capacity limits and temporal feasibility at every stop.

In commercial freight operations, the optimization objective is strictly lexicographic: dispatchers prioritize minimizing the total number of deployed vehicles (N_V) before minimizing the total travel distance (D_T) [3], [4]. Formally, letting $\mathcal{S}$ denote the space of feasible routing solutions, a solution $s_1 \in \mathcal{S}$ is strictly preferred over $s_2 \in \mathcal{S}$ (denoted $s_1 \prec s_2$) if and only if:

$$NV(s_1) < NV(s_2) \quad \vee \\ (NV(s_1) = NV(s_2) \wedge TD(s_1) < TD(s_2)) \quad (1)$$

This hierarchical ordering reflects the economic structure of commercial transportation, where fixed vehicle capital acquisition, maintenance, insurance, and driver shift wages dominate marginal fuel consumption.

Adaptive Large Neighborhood Search (ALNS) [5], [6] has long served as a leading metaheuristic paradigm for VRPTW, exploring diverse neighborhood structures via iterative ruin-and-recreate (destroy and repair) operations while strictly preserving feasibility. However, standard ALNS implementations rely on a stateless roulette-wheel or multi-armed bandit mechanism that updates selection probabilities from aggregated historical reward averages. This design exhibits two structural weaknesses:

- 1) **Stateless Operator Selection:** Heuristic selection weights depend solely on historical scores across coarse iteration segments, ignoring the instantaneous search state, such as spatial route fragmentation, remaining temporal slack, or capacity utilization.
- 2) **Myopic Plateau Stagnation:** Classical selection mechanisms struggle to coordinate multi-step operator sequences required to escape structural plateaus, frequently stalling when fleet-size reduction requires evaluating temporarily non-improving intermediate states.

In parallel, Neural Combinatorial Optimization (NCO) has emerged as a data-driven alternative, using Deep Reinforcement Learning (DRL) models such as Pointer Networks [7], Attention Models [8], and reinforcement learning frameworks for routing [9]. While these approaches capture spatial regularities without manual heuristic engineering, direct application of pure end-to-end neural solvers to industrial VRPTW encounters significant challenges:

- 1) **Feasibility Degradation under Tight Time Windows:** Autoregressive decoding policies often face constraint conflicts when customer time windows are narrow, leading to high constraint violation rates or requiring surrogate penalty tunings [10], [11].
- 2) **Objective Misalignment:** Most neural formulations optimize scalarized travel costs rather than strictly

enforcing the primary-secondary fleet minimization hierarchy ($N_V \succ D_T$).

- 3) **Out-of-Distribution Sensitivity:** Pure constructive policies trained on synthetic uniform instances often experience performance degradation when applied to larger instances or heterogeneous spatial topologies without extensive fine-tuning.

To bridge these paradigms, this paper presents Tri-Level Hybrid DDQN-ALNS, a learning-augmented optimization framework that embeds state-conditioned reinforcement learning directly inside the iterative search loop of ALNS. Rather than replacing the metaheuristic with an opaque black-box neural solver, our approach maintains hard operational constraint feasibility and interpretable neighborhood operators from operations research, structuring search governance into three coordinated tiers: two learned reinforcement learning tiers (Macro Plateau Governance and Micro Operator Selection) coupled with a high-throughput heuristic execution tier.

The primary contributions of this paper are summarized as follows:

- **Three-Tier Hierarchical Control Architecture:** We formulate metaheuristic search management as a Semi-Markov Decision Process over an Aggregated Feature Space (Aggregated SMDP), decoupling macro-level search exploration regimes (Plateau Controller) from micro-level destroy-repair operator selection (Operator Controller).
- **State-Conditioned Micro Selection with Policy-Concentration Gating:** We develop a Dueling Double Deep Q-Network (DDQN) operating over a 20-dimensional spatiotemporal state space, incorporating a normalized Softmax-Entropy policy-concentration gate to adaptively blend learned neural Q-values with empirical variance-aware UCB bandit priors.
- **Learned Acceptance Criterion (LAC) as Predictive Override:** We introduce a supervised neural acceptance model trained on delayed downstream lookahead signals ($H_{lac} = 80$), augmenting Simulated Annealing with state-aware predictive lookahead to rescue trapped exploratory trajectories during deep cooling without vetoing valid SA acceptances.
- **Spatial GNN Edge Guidance and High-Throughput Search:** We combine an offline-trained Contrastive Graph Neural Network (GNN) and granular candidate restriction that prunes 97.5% – 98.7% of unpromising insertion arcs with $O(1)$ forward-slack feasibility testing and time-limited Set Partitioning integer programming recombination via HiGHS.
- **Rigorous Scale-Aware Benchmark across 74 Instances:** We evaluate the framework across 74 standard benchmark instances (56 Solomon-100, 12 Homberger-200, and 6 Homberger-400) under strictly isolated independent cold-starts, providing an empirical characterization of solution quality, fleet reduction, and computa-

tional trade-offs supported by non-parametric statistical hypothesis testing.

The remainder of this paper is organized as follows: Section II reviews related literature. Section III presents the formal problem formulation. Section IV describes the proposed Tri-Level Hybrid DDQN-ALNS methodology. Section V details the empirical benchmark results, statistical tests, and ablation analyses. Section VI discusses scale-aware dynamics, analytical properties, and limitations. Finally, Section VII concludes the paper and outlines future directions.

II. RELATED WORK

A. CLASSICAL METAHEURISTICS AND ALNS FOR VRPTW

Metaheuristics have long dominated the practical resolution of the VRPTW [4]. Building on the seminal insertion heuristics of Solomon [2] and the Large Neighborhood Search (LNS) paradigm of Shaw [17], Ropke and Pisinger [5], [6], [18] formalized the Adaptive Large Neighborhood Search (ALNS) framework, in which a portfolio of handcrafted removal and insertion operators competes for selection via roulette-wheel probabilities updated from historical performance scores.

Subsequent research advanced two complementary directions. For the hierarchical objective of fleet minimization, Guided Ejection Search [19] and penalty-based edge assembly memetic algorithms [20] remain the strongest dedicated mechanisms. In parallel, matheuristic hybrids exploit an elite route pool—tracing back to the adaptive memory of Rochat and Taillard [21]—and periodically solve a Set Partitioning model over accumulated routes [13], [22], [23], recombining high-quality route fragments beyond the reach of local moves. More recently, Slack Induction by String Removals [12] demonstrated that carefully designed destroy operators alone can rival far more complex frameworks, underscoring how decisive the destroy step is.

Despite this empirical success, the adaptive layer of ALNS exhibits an intrinsic architectural limitation. Operator weights are updated as exponentially smoothed averages over coarse segments of η iterations, yielding credit assignment that is both delayed and aggregated; critically, the resulting selection distribution is stateless, conditioning on neither the spatial topology of the incumbent (e.g., cluster structure, inter-route proximity) nor its temporal characteristics (e.g., time-window tightness, slack distribution). Performance is further sensitive to manually tuned hyperparameters such as the decay factor and reward scores. Notably, a recent meta-analysis by Turkeš et al. [24] quantified that this adaptive mechanism contributes only marginal gains over non-adaptive LNS—strong evidence that the operator-selection policy, rather than the operator set itself, is the underexploited component of the framework.

B. NEURAL COMBINATORIAL OPTIMIZATION AND END-TO-END DEEP RL

In parallel with classical heuristics, Neural Combinatorial Optimization (NCO) has gained widespread traction by learning data-driven routing policies. Originating from Pointer Networks [7] and policy-gradient formulations for the Traveling Salesperson Problem [25], Nazari et al. [9] extended end-to-end reinforcement learning to the Capacitated VRP by decoupling static customer coordinates from dynamic demand states. Subsequently, Kool et al. [8] established the Attention Model (AM) as the prevailing benchmark, leveraging Transformer-style self-attention with a greedy rollout baseline. Graph Neural Networks (GNNs) [26], [27] have likewise been deployed to capture permutation-invariant spatial graph embeddings.

More recently, several architectures have attempted direct end-to-end construction for VRPTW [10], [11]. Despite their impressive inference speed on small-scale benchmark instances, pure end-to-end neural solvers face four fundamental bottlenecks that impede their deployment in industrial VRPTW settings:

- 1) **Feasibility Fragility under Tight Time Windows:** While capacity constraints can be masked during autoregressive decoding, hard time windows frequently induce structural deadlocks where no remaining unvisited customer is temporally reachable, forcing the constructive policy to spawn redundant vehicles or incur severe penalty violations.
- 2) **Lexicographic Misalignment:** Pure DRL frameworks are trained to minimize a scalarized objective (typically travel distance or a weighted surrogate), lacking the mechanism to strictly enforce the lexicographic priority ($N_V \succ D_T$) that governs fleet economics.
- 3) **Out-of-Distribution (OOD) Scalability Collapse:** Models trained on synthetic uniform distributions at $N \in \{50, 100\}$ exhibit steep performance degradation when evaluated on larger problem instances ($N \in \{200, 400\}$) or heterogeneous spatial clusters without expensive instance-specific fine-tuning.
- 4) **Lack of Operational Auditability:** Constructing an entire routing plan via end-to-end neural forward passes functions as an opaque black box, offering no interpretable intermediate steps for dispatchers to audit, constrain, or manually adjust in real-time dispatch systems.

C. LEARNING-AUGMENTED METAHEURISTICS

To circumvent the feasibility and scalability bottlenecks of pure neural construction, the “Learning to Optimize” paradigm [28] embeds machine learning models directly within proven metaheuristic frameworks. Rather than generating solutions from scratch, learning agents guide critical decision points inside iterative search loops, such as operator selection [14], [29], neural solution perturbation [30], and learned destroy policies [15], [16].

TABLE 1. Taxonomy of Representative Methodologies for VRPTW and Related Routing Problems

Methodology Category	Representative Works	Hard TW Feasible	Objective Paradigm	State-Aware Selection	Hierarchical Control	Learned Accept	Per-Step Compute Cost
Classical Metaheuristics	ALNS [5], [6]	✓	Lexicographic	× (Stateless)	×	× (SA)	Minimal (Heuristic)
	SISR [12], HGS [13]	✓	Lexicographic	× (Rule-based)	×	× (SA/Thresh)	Minimal (Heuristic)
End-to-End Neural Solvers	AM [8], DRL-VRP [9]	× (Penalized)	Scalarized	N/A	×	N/A	Very Low (One-shot)
	Neural VRPTW [10], [11]	× (Mask-fragile)	Scalarized	N/A	×	N/A	Very Low (One-shot)
Single-Agent Hybrid RL-LNS	Learned LNS [14], [15]	✓	Scalarized	✓	× (Single-agent)	× (SA)	Moderate (Policy NN)
	Neural LNS [16]	✓	Scalarized	✓	× (Single-agent)	× (Greedy)	Moderate (Policy NN)
Proposed Tri-Level Hybrid DDQN-ALNS	This Work	✓	Lexicographic	✓	✓ (Dual HMDP)	✓ (LAC)	High (1.5–4× RL/MIP)

While these hybrid frameworks preserve the underlying validity of metaheuristic search, existing learning-augmented approaches for routing exhibit three key shortcomings:

- 1) **Single-Level Decision Architecture:** Prior RL-augmented LNS methods deploy a flat, single-agent policy that selects operators uniformly across all search stages. They lack hierarchical decoupling between macro-level exploration regimes (e.g., detecting search stagnation and triggering aggressive plateau escape) and micro-level operator selection conditioned on local topology.
- 2) **Persistence of Static Acceptance Criteria:** Despite deploying neural operator selectors, existing hybrids continue to rely on conventional Simulated Annealing with static geometric cooling schedules. Consequently, candidate moves are judged solely by instantaneous cost degradation ΔC , ignoring structural potential that could unlock downstream improvements.
- 3) **Absence of Lexicographic Multi-Objective Coordination:** Most learned metaheuristics focus on scalarized objectives (e.g., distance minimization in CVRP), without architectural mechanisms designed for the strict lexicographic priority ($N_V \succ D_T$) fundamental to VRPTW fleet operations.

D. POSITIONING OF THIS WORK

Table 1 situates the proposed Tri-Level Hybrid DDQN-ALNS framework within the landscape of vehicle routing methodologies. Our approach distinctively combines the hard constraint guarantees of classical operations research with a dual-agent Hierarchical MDP and a Learned Acceptance Criterion, providing a rigorous, state-aware, and lexicographically aligned optimization engine. Critically, Table 1 also makes explicit the operational trade-off: while classical heuristics operate at raw, stateless evaluation speeds (≈ 200 – 450 it/s) and end-to-end neural architectures prioritize sub-second inference at the expense of feasibility, our

framework deliberately incurs higher per-iteration computational overhead ($1.5 \times$ – $4 \times$ per-step slowdown due to online RL experience updates, feature extraction, and periodic MIP recombination) to achieve superior asymptotic search quality and escape combinatorial stagnation.

Unlike single-agent RL-LNS frameworks that condition operator selection on a single flattened state vector, macro/micro decoupling addresses a credit-assignment problem across timescales: reward for escaping a multi-hundred-iteration plateau is temporally distant from any individual destroy-repair decision, and folding both signals into one policy forces a single network to learn fast local corrections and slow structural regime shifts from the same gradient. The macro tier’s semi-MDP formulation over $\Delta_{\text{seg}} = 100$ -iteration options [43] isolates plateau-detection credit assignment from micro-level operator credit assignment, letting each network specialize to its own effective time horizon. We do not claim this decoupling is superior in the abstract; the ablation in Section V-F disabling the Macro Plateau Controller is the empirical test, and the resulting variance inflation at $N = 200$ is the evidence, not the architecture itself.

III. PROBLEM FORMULATION

The Vehicle Routing Problem with Time Windows (VRPTW) is formally defined on a complete directed graph $\mathcal{G} = (\mathcal{V}, \mathcal{A})$. The vertex set $\mathcal{V} = \mathcal{N} \cup \{0, n+1\}$ comprises customer nodes $\mathcal{N} = \{1, 2, \dots, n\}$, origin depot node 0, and destination depot node $n+1$. The physical depot is modeled as two separate nodes (0 and $n+1$) to represent route departure and return without self-loops. The arc set is $\mathcal{A} = \{(i, j) : i, j \in \mathcal{V}, i \neq j, i \neq n+1, j \neq 0\}$.

Each customer $i \in \mathcal{N}$ has a non-negative demand $q_i > 0$, service duration $s_i \geq 0$, and service time window $[e_i, l_i]$, where e_i and l_i denote the earliest and latest allowable service start times, respectively. Depot nodes satisfy $q_0 = q_{n+1} = 0$, $s_0 = s_{n+1} = 0$, and $[e_0, l_{n+1}]$ defines the global planning horizon. Each arc $(i, j) \in \mathcal{A}$ has travel distance $c_{ij} \geq 0$

and travel time $t_{ij} \geq 0$. A homogeneous fleet of K vehicles, denoted by $\mathcal{K} = \{1, \dots, K\}$, is stationed at the depot, with each vehicle possessing carrying capacity Q .

A. DECISION VARIABLES

The optimization model employs two sets of decision variables:

- $x_{ij}^k \in \{0, 1\}$: Binary variable equal to 1 if vehicle $k \in \mathcal{K}$ traverses arc $(i, j) \in \mathcal{A}$, and 0 otherwise.
- $w_i^k \geq 0$: Continuous variable specifying the time at which vehicle $k \in \mathcal{K}$ begins service at node $i \in \mathcal{V}$.

B. MATHEMATICAL MODEL

The primary objective is lexicographic: first minimizing the number of deployed vehicles (N_V), and secondarily minimizing total travel distance (D_T):

$$\text{lex min} \left(\sum_{k \in \mathcal{K}} \sum_{j \in \mathcal{N}} x_{0j}^k, \sum_{k \in \mathcal{K}} \sum_{(i,j) \in \mathcal{A}} c_{ij} x_{ij}^k \right) \quad (2)$$

In computational solver implementations, this hierarchy is evaluated via a scalarized surrogate objective using a large penalty $\mathcal{M} \gg \sum_{(i,j) \in \mathcal{A}} c_{ij}$:

$$\min \sum_{k \in \mathcal{K}} \left(\mathcal{M} \sum_{j \in \mathcal{N}} x_{0j}^k + \sum_{(i,j) \in \mathcal{A}} c_{ij} x_{ij}^k \right) \quad (3)$$

subject to:

$$\sum_{k \in \mathcal{K}} \sum_{j \in \mathcal{V}, j \neq i} x_{ij}^k = 1, \quad \forall i \in \mathcal{N} \quad (4)$$

$$\sum_{j \in \mathcal{N} \cup \{n+1\}} x_{0j}^k \leq 1, \quad \forall k \in \mathcal{K} \quad (5)$$

$$\sum_{j \in \mathcal{N} \cup \{0\}} x_{j,n+1}^k - \sum_{j \in \mathcal{N} \cup \{n+1\}} x_{0j}^k = 0, \quad \forall k \in \mathcal{K} \quad (6)$$

$$\sum_{i \in \mathcal{V}, i \neq h} x_{ih}^k - \sum_{j \in \mathcal{V}, j \neq h} x_{hj}^k = 0, \quad \forall h \in \mathcal{N}, k \in \mathcal{K} \quad (7)$$

$$\sum_{i \in \mathcal{N}} q_i \sum_{j \in \mathcal{V}, j \neq i} x_{ij}^k \leq Q, \quad \forall k \in \mathcal{K} \quad (8)$$

$$w_i^k + s_i + t_{ij} - M_{ij}(1 - x_{ij}^k) \leq w_j^k, \quad \forall (i, j) \in \mathcal{A}, k \in \mathcal{K} \quad (9)$$

$$e_i \leq w_i^k \leq l_i, \quad \forall i \in \mathcal{V}, k \in \mathcal{K} \quad (10)$$

$$x_{ij}^k \in \{0, 1\}, \quad w_i^k \geq 0, \quad \forall (i, j) \in \mathcal{A}, k \in \mathcal{K} \quad (11)$$

Constraint (4) ensures that every customer is visited exactly once. Constraints (5) and (6) enforce that each vehicle departs from origin depot 0 and arrives at destination depot $n+1$ at most once. Flow conservation across all customer nodes is guaranteed by (7). Constraint (8) prevents vehicle capacity violations. Schedule propagation is maintained via (9), where $M_{ij} = \max\{0, l_i + s_i + t_{ij} - e_j\}$ defines the

tightest valid Big- M upper bound. Customer time-window compliance is governed by (10); if a vehicle arrives at customer i prior to e_i , it incurs an idle waiting time such that service begins at $w_i^k = e_i$. Finally, (11) defines the variable domains. Note that while w_i^k is indexed per vehicle for strict MIP formulation alignment with x_{ij}^k , constraint (4) ensures that exactly one vehicle serves each customer i , rendering only one w_i^k active in any feasible dispatch plan.

IV. PROPOSED METHODOLOGY

This section details the proposed Tri-Level Hybrid DDQN-ALNS framework. The architecture integrates online adaptive reinforcement learning decision layers with operations research search mechanics to optimize the lexicographic VRPTW objective while maintaining operational constraint feasibility.

A. TRI-LEVEL ARCHITECTURE OVERVIEW

The framework decomposes the global routing search into three coordinated operational tiers, illustrated in Fig. 1:

- 1) **Macro Decision Tier (Plateau Controller π_{macro}):** Formulated as a semi-Markov Decision Process (semi-MDP) operating over iteration segments of length $\Delta_{\text{seg}} = 100$. A dedicated Dueling Double Deep Q-Network (Macro DDQN) [40], [41] with Prioritized Experience Replay [42] evaluates convergence velocity, detects stagnation, and transitions among 7 strategic exploration and recombination regimes.
- 2) **Micro Selection Tier (Operator Controller π_{micro} & LAC):** Operates at each individual search iteration t . A Micro Dueling DDQN with normalized Softmax-Entropy policy-concentration gating selects composite destroy-repair operator pairings conditioned on a 20-dimensional spatiotemporal state vector. A Learned Acceptance Criterion (LAC) augments Simulated Annealing by evaluating candidate acceptance based on delayed downstream improvement promise.
- 3) **Heuristic Search Tier (Granular ALNS & Route-Pool Recombination):** Executes ruin-and-recreate operations, accelerates feasibility testing via $O(1)$ Savelsbergh forward time slack checks per candidate insertion, restricts local moves to top- k ($k = 25$) spatiotemporally correlated candidate neighbors, and periodically solves a time-limited Set Partitioning formulation over accumulated elite routes via HiGHS (4.0s wall-clock cutoff).

B. MACRO LEVEL: PLATEAU CONTROLLER (AGGREGATED SMDP FORMULATION)

We formulate the macro search governance tier as a Semi-Markov Decision Process [43] over an Aggregated State Representation (Aggregated SMDP) defined by the tuple $(\mathcal{S}_{\text{macro}}, \mathcal{A}_{\text{macro}}, P_{\text{macro}}, R_{\text{macro}}, \gamma_{\text{macro}})$. Because the underlying combinatorial route pool and fine-grained trajectory history are projected into a finite 13-dimensional feature space $\mathbf{s}_k^{\text{macro}}$, the decision tier operates over a compact aggregated

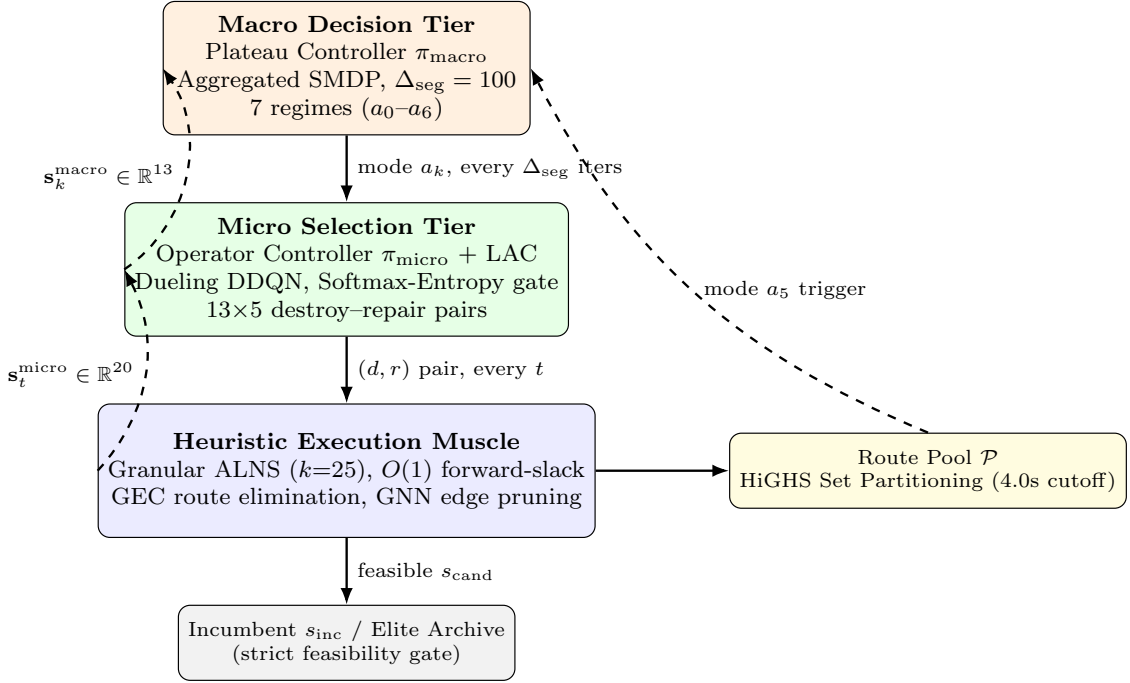


FIGURE 1. Tri-Level Hybrid DDQN-ALNS control architecture. Solid arrows: downward control flow. Dashed arrows: upward state feedback consumed by the RL training loops.

state representation with temporal options of fixed duration $\Delta_{seg} = 100$ low-level ALNS iterations. The temporally extended transition kernel $P_{macro}(\mathbf{s}_{k+1}^{macro} | \mathbf{s}_k^{macro}, a_k)$ represents the cumulative transition distribution over the 100 underlying stochastic destroy-repair steps, with effective discount factor $\gamma_{macro} = \gamma^{\Delta_{seg}}$.

1) Macro State Representation

At the start of segment k , the macro controller observes a 13-dimensional state vector $\mathbf{s}_k^{macro} \in \mathbb{R}^{13}$, capturing convergence velocity, fleet economics, and solution topology:

$$\mathbf{s}_k^{macro} = \left[\min\left(\frac{\text{no_imp}}{\text{patience}}, 1\right), \min\left(\frac{C_{cur} - C_{best}}{C_{best}}, 1\right), \min\left(\frac{T}{T_0}, 1.5\right), \text{imp_rate}, \min\left(\frac{NV_{cur}}{NV_{init}}, 2\right), \frac{NV_{cur} - NV_{best}}{\max(NV_{init}, 1)}, rb, lb, \tau_{tight}, \bar{s}, \rho_{fill}, \rho_{pool}, \frac{k}{K_{max}} \right] \quad (12)$$

where no_imp is the number of consecutive unimproved iterations, T/T_0 is the normalized annealing temperature, $rb \in [0, 1]$ and $lb \in [0, 1]$ measure spatial route separation and capacity balance, $\tau_{tight} = \frac{1}{n} \sum_{i=1}^n \mathbb{I}(l_i - e_i < 0.5 \cdot T_{depot})$ represents the fraction of tight time-window customers relative to the depot operational horizon $T_{depot} = l_{n+1} - e_0$, $\bar{s}$ is average forward time slack, ρ_{fill} is average vehicle capacity utilization, and $\rho_{pool} = \min(|\mathcal{P}|/\mathcal{P}_{max}, 1.0)$ tracks route pool fill ratio.

2) Macro Action Space & Search Regimes

The macro action space $\mathcal{A}_{macro}$ comprises 7 distinct search modes ($|\mathcal{A}_{macro}| = 7$):

- a_0 (DEFAULT): Balanced exploration under standard parameters.
- a_1 (INTENSE_EXPLOIT): Deep cooling with intensified multi-pass local search ($\gamma_{destroy} \leftarrow 0.7\gamma_0$).
- a_2 (DIVERSIFY): Thermal reheating ($T \leftarrow 1.5T$) paired with high destroy ratios ($\gamma_{destroy} \leftarrow 1.5\gamma_0$).
- a_3 (TW_RESCUE): Targeted ruin-and-recreate prioritizing tight time-window customer clusters with forward slack below the 20th percentile.
- a_4 (ROUTE_ELIMINATE): Biased destruction targeting the shortest routes via Generalized Ejection Chains.
- a_5 (POOL_RECOMBINE): Time-limited Set Partitioning integer programming recombination across elite routes in $\mathcal{P}$ via HiGHS branch-and-cut (4.0s cutoff).
- a_6 (INFEASIBLE_DESCENT): Controlled temporal and capacity violation relaxation governed by an adaptive Lagrangian penalty objective $C_{pen}(s) = C(s) + \lambda_{tw} \sum_i \max(0, a_i - l_i) + \lambda_{cap} \sum_r \max(0, q(r) - Q)$, where multipliers $\lambda_{tw}, \lambda_{cap} \in [0.1, 100.0]$ adapt every 10 iterations according to $\lambda \leftarrow \min(\lambda \cdot 1.2, 100.0)$ if infeasible move ratio exceeds 50%, and $\lambda \leftarrow \max(\lambda/1.2, 0.1)$ otherwise.

3) Multi-Objective Segment Reward & Surrogate Policy Optimization

To guide reinforcement learning exploration in alignment with the primary-secondary optimization hierarchy, the seg-

ment reward R_{seg} balances vehicle reduction gains against distance savings via an adaptive fleet pressure factor $\lambda_{\text{fleet}} \in [0, 1]$:

$$R_{\text{seg}} = \text{Base} + \lambda_{\text{fleet}} \Psi_{NV} + (1 - \lambda_{\text{fleet}}) \Psi_{TD} + \text{Bonus}_{NV} + \Delta\Phi \quad (13)$$

where $\Psi_{NV} = \sum_{j=1}^4 \beta_{NV,j} \Delta X_j$ and $\Psi_{TD} = \sum_{j=1}^4 \beta_{TD,j} \Delta X_j$ linearly weight improvements across best-so-far and incumbent fleet size (ΔNV) and travel distance (ΔC), with representative primary fleet coefficient $\beta_{NV,1} = 8.0$. Baseline computational penalty Base penalizes intensive local search passes and unproductive stagnation, while Bonus_{NV} provides discrete milestone reinforcement (+15.0) upon physical vehicle elimination. Complete parameter listings and definitions are tabulated in Supplementary Material Table S1.

The presence of two parallel fleet-reward pathways in (13) addresses a fundamental credit-assignment challenge across temporal scales:

- 1) **Dense Continuous Guidance** ($\lambda_{\text{fleet}} \Psi_{NV}$): Operates as a dense, incremental reward signal across every macro segment, encouraging moves that balance inter-route capacity utilization and narrow the vehicle count gap relative to the incumbent.
- 2) **Sparse Milestone Reinforcement** (Bonus_{NV}): Serves as a high-magnitude discrete milestone bonus (+15.0) awarded exclusively when a physical vehicle elimination actually occurs, preventing the sparse route collapse signal from being diluted across routine distance variations.

a: Experimental Firewall and Reward Calibration

All reward constants in (13)–(15) and structural parameters were calibrated on independent synthetic pilot topologies generated via domain randomization during initial prototyping (detailed in Supplementary Material Section 9 and Table S1). To prevent data leakage and hyperparameter overfitting, a strict experimental firewall was enforced: zero reward tuning, grid search, or post-hoc parameter adjustments were conducted on any of the 74 benchmark test instances (56 Solomon-100, 18 Gehring-Homberger). All test instances were evaluated strictly out-of-sample under isolated cold-starts.

We emphasize a fundamental architectural dichotomy between the combinatorial evaluation layer and the reinforcement learning training layer:

- 1) **Strict Lexicographic Evaluation**: Incumbent replacement, elite archive admission, and final solution selection strictly evaluate the true lexicographic ordering $f(s) = (NV(s), TD(s))$, guaranteeing that fleet reduction strictly dominates distance.
- 2) **Smooth RL Reward Surrogate**: The reinforcement learning policy receives scalarized reward feedback R_{seg} modulated by λ_{fleet} . When surplus capacity exists ($NV_{\text{cur}} > NV_{\text{best}}$), setting $\lambda_{\text{fleet}} \rightarrow 1$ concentrates gradients on fleet collapse; once fleet size stabilizes

($NV_{\text{cur}} = NV_{\text{best}}$), setting $\lambda_{\text{fleet}} \leq 0.5$ transfers gradient sensitivity to routing distance.

To accelerate learning while preventing artificial reward cycles in the surrogate MDP, we incorporate potential-based reward shaping $\Delta\Phi = \gamma_{\text{macro}} \Phi(s_{k+1}^{\text{macro}}) - \Phi(s_k^{\text{macro}})$. We explicitly note that this term implements potential-based shaping of the macro surrogate objective [31] rather than guaranteeing policy invariance with respect to the underlying lexicographic VRPTW objective (which is enforced externally by the combinatorial evaluator). The adaptive fleet pressure factor $\lambda_{\text{fleet}} \in [0, 1]$ dynamically shifts optimization priority toward fleet reduction whenever surplus capacity exists:

$$\lambda_{\text{fleet}}(s) = \sigma\left(8.0 \cdot \frac{NV_{\text{cur}} - NV_{\text{best}}}{NV_{\text{init}}}\right) \quad (14)$$

where $\sigma(\cdot)$ is the logistic sigmoid, with the state potential defined as:

$$\begin{aligned} \Phi(s) = & -\lambda_{\text{fleet}}(s) \frac{\max(NV - NV_{\text{best}}, 0)}{NV_{\text{init}}} \\ & - (1 - \lambda_{\text{fleet}}(s)) \cdot 0.5 \cdot \text{clip}\left(\frac{TD - TD_{\text{best}}}{TD_{\text{best}}} \cdot 100, -25, 25\right) \end{aligned} \quad (15)$$

The Macro Plateau Controller trains a dedicated Dueling Double DQN network (Q_{macro}) with prioritized replay buffer $\mathcal{D}_{\text{macro}}$, executing Adam gradient steps with Cosine Annealing learning rate scheduling at the completion of each macro segment.

C. MICRO LEVEL: OPERATOR CONTROLLER & ACTION BLENDING

1) Micro State Representation

At each iteration t , the operator controller observes a 20-dimensional state vector $s_t^{\text{micro}} \in \mathbb{R}^{20}$ (Table 2), detailing fine-grained search dynamics and route topology.

2) Composite Action Space

The micro action space $\mathcal{A}_{\text{micro}}$ comprises $N_D \times N_R = 13 \times 5 = 65$ composite heuristic pairs:

- **13 Destroy Operators** ($N_D = 13$): Random, Worst-cost, Shaw spatiotemporal relatedness, Route portion removal, Time-window urgency removal, Route-Depletion Destroy, Route dispersion elimination, Costly route elimination, Cross-route Shaw, Route merge sampling, Route absorb disruption, Neural worst, and Neural Shaw.
- **5 Repair Operators** ($N_R = 5$): Greedy insertion, 2-Regret, 3-Regret, Time-window greedy, and Forward-time-slack greedy (FTS-Greedy).

3) Dueling DDQN Architecture & Policy-Concentration Blending

The Micro Q-network adopts a Dueling architecture [32] with layer normalization:

$$Q(s, a; \theta) = V(s; \theta_v) + \left(A(s, a; \theta_a) - \frac{1}{|\mathcal{A}|} \sum_{a'} A(s, a'; \theta_a) \right) \quad (16)$$

To adaptively balance learned neural predictions with exploration priors, we implement a normalized Softmax-Entropy Policy-Concentration Gate. Let $\pi_\theta(a | s) = \text{softmax}(Q(s, a)/\tau_{\text{soft}})$ be the Boltzmann policy at softmax temperature $\tau_{\text{soft}} = 1.0$. The normalized Shannon entropy is defined as:

$$\mathcal{H}(\pi_\theta(\cdot | s)) = -\frac{1}{\ln |\mathcal{A}|} \sum_{a \in \mathcal{A}} \pi_\theta(a | s) \ln \pi_\theta(a | s) \in [0, 1] \quad (17)$$

We define the policy-concentration weight as $w_{\text{conf}}(s) = 1.0 - \mathcal{H}(\pi_\theta(\cdot | s))$. High concentration ($w_{\text{conf}} \rightarrow 1$) indicates that the Boltzmann policy is sharply peaked around a dominant operator pair, whereas low concentration ($w_{\text{conf}} \rightarrow 0$) signals an exploratory regime.

To ensure commensurability across diverse score scales, components are standardized: (1) neural Q-values are z-score standardized, $\tilde{Q}(s, a) = (Q(s, a) - \bar{Q}(s))/(\sigma_Q(s) + \epsilon)$, with instantaneous mean $\bar{Q}(s)$ and standard deviation $\sigma_Q(s)$; (2) heuristic log-priors are min-max normalized, $\tilde{P}(a) \in [0, 1]$; and (3) historical bandit rewards $\bar{\mu}_{\text{bandit}}(a)$ are tracked via Welford updates. The effective action-selection score

TABLE 2. Feature Definition of the 20-Dimensional Micro State s_t^{micro}

Index	Feature Symbol	Mathematical Definition / Description
s_1	ΔC_{rel}	$\min((C_{\text{cur}} - C_{\text{best}}) / \max(C_{\text{best}}, 1), 1.0)$
s_2	Ratio_{NV}	$\min(NV_{\text{cur}} / \max(NV_{\text{init}}, 1), 2.0)$
s_3	$\text{Prog}_{\text{global}}$	$t / \max(t_{\text{max}}, 1)$
s_4	Prog_{seg}	$(t \bmod \Delta_{\text{seg}}) / \Delta_{\text{seg}}$
s_5	T_{norm}	$\min(T / \max(T_0, 10^{-6}), 1.5)$
s_6	$\text{Stag}_{\text{ratio}}$	$\min(\text{no_imp} / \max(\text{patience}, 1), 1.0)$
s_7	rb	Inter-route spatial separation index $\in [0, 1]$
s_8	lb	Vehicle capacity load balance metric $\in [0, 1]$
s_9	τ_{tight}	Fraction of customers with tight time windows
s_{10}	$\bar{s}$	Normalized average forward time slack $\in [0, 1]$
s_{11}	ρ_{fill}	Fleet-wide capacity utilization ratio $\in [0, 1]$
s_{12}	ρ_{pool}	Route pool fill ratio $\min(\mathcal{P} / \mathcal{P}_{\text{max}}, 1.0)$
s_{13}	m_{idx}	Active macro mode index $\text{mode_idx} / (\mathcal{A}_{\text{macro}} - 1)$
s_{14}	ΔNV_{rel}	$(NV_{\text{cur}} - NV_{\text{best}}) / \max(NV_{\text{init}}, 1)$
s_{15}	Imp_{rec}	Exponentially smoothed recent improvement rate
s_{16}	Gini_{len}	Gini coefficient of route customer lengths
s_{17}	μ_{slack}	Empirical mean of customer forward time slacks
s_{18}	σ_{slack}	Standard deviation of forward time slacks
s_{19}	Frac_{cap}	Fraction of routes exceeding 90% capacity
s_{20}	Var_{dist}	Variance of inter-route centroid distances

$Q_{\text{eff}}(s, a)$ dynamically blends neural Q-values with historical priors and the variance-aware UCB bonus adapted from UCB [39] and UCB-V [38], with the exploration bonus explicitly gated by $(1 - w_{\text{conf}}(s))$:

$$Q_{\text{eff}}(s, a) = w_{\text{conf}}(s) \tilde{Q}(s, a) + (1 - w_{\text{conf}}(s)) \left[\alpha_p \tilde{P}(a) + \beta_b \bar{\mu}_{\text{bandit}}(a) + c_{\text{ucb}} \sigma_a \sqrt{\ln N_t / N_a(t)} \right] \quad (18)$$

where $\alpha_p = 0.35$ scales prior log-probabilities, $\beta_b = 0.65$ weights decaying historical posterior rewards $\bar{\mu}_{\text{bandit}}(a)$, $c_{\text{ucb}} = 1.0$ governs the UCB exploration bonus, and $\sigma_a = \sqrt{M_{2,a} / \max(N_a(t) - 1, 1)}$ denotes the online sample standard deviation tracked via Welford's algorithm. Action counters are initialized with pseudocount $N_a(0) = 1$ ($\forall a \in \mathcal{A}$) and $N_t = \sum_a N_a(t)$ for finite numerical bounds.

Gating priors and UCB bonuses by $(1 - w_{\text{conf}}(s))$ ensures that as policy concentration approaches 1 ($w_{\text{conf}} \rightarrow 1$), selection delegates to learned neural Q-values $\tilde{Q}(s, a)$, while smoothly reverting to variance-aware UCB exploration when policy entropy is diffuse ($w_{\text{conf}} \rightarrow 0$).

D. LEARNED ACCEPTANCE CRITERION (LAC): SA WITH PREDICTIVE LOOKAHEAD OVERRIDE

In classical ALNS, move acceptance relies on Simulated Annealing (SA), which accepts improving transitions ($\Delta C < 0$) deterministically and non-improving transitions ($\Delta C \geq 0$) with probability $P_{\text{SA}} = \exp(-\Delta C/T)$. During deep cooling stages ($T \rightarrow 0$), P_{SA} vanishes exponentially, restricting search to downhill moves and causing premature local trapping.

To prevent search entrapment without compromising the convergence stability of Simulated Annealing, the Learned Acceptance Criterion (LAC) operates as an online supervised binary classifier $\phi_{\text{LAC}} : \mathbb{R}^9 \rightarrow [0, 1]$. Rather than serving as an exclusionary filter or replacement, LAC implements a disjunctive acceptance decision rule:

$$\text{Accept}(s' | s, T) \iff (p_t \geq \tau_{\text{acc}}(t)) \vee (P_{\text{SA}}(\Delta C, T) > U) \quad (19)$$

where $p_t = \phi_{\text{LAC}}(\mathbf{z}_t)$, $U \sim \text{Uniform}(0, 1)$, and $\tau_{\text{acc}}(t) \in [0.40, 0.75]$ is an adaptive confidence threshold.

Crucially, the disjunctive operator ($\vee$) guarantees that LAC acts strictly as an **acceptance booster / rescue mechanism** via two properties: (i) **Preservation of Downhill Transitions**: Any move accepted by standard Simulated Annealing ($P_{\text{SA}} > U$) is unconditionally accepted, preserving classical asymptotic convergence guarantees; and (ii) **Predictive Rescue of Exploratory Transitions**: When Simulated Annealing rejects a non-improving move ($P_{\text{SA}} \leq U$) during cold cooling stages, LAC evaluates whether the topological structural transition is likely to yield a downstream incumbent improvement within lookahead horizon $H_{\text{lac}} = 80$ iterations. If predicted probability exceeds the threshold ($p_t \geq \tau_{\text{acc}}$), the move is accepted as an exploratory rescue step.



Because decision rule (19) operates as an additive booster, it systematically increases the volume of accepted non-improving transitions during deep cooling. In our ablation and limitations analyses (Sections V-F and VI-D), we discuss the methodological implication of distinguishing state-conditioned lookahead intelligence from generic acceptance rate inflation.

1) Feature Representation and Endogenous Training Dynamics

The LAC input feature vector $\mathbf{z}_t \in \mathbb{R}^9$ comprises 9 normalized features:

$$\mathbf{z}_t = \left[\text{sign}(\Delta C) \sqrt{|\Delta C|/C_{\text{best}}}, \frac{T}{T_0}, \frac{\text{no_imp}}{\text{patience}}, \frac{NV_{\text{cur}}}{NV_{\text{init}}}, \Delta NV, \rho_{\text{fill}}, \bar{s}, \frac{t}{T_{\text{max}}}, \exp\left(-\frac{\max(\Delta C, 0)}{T}\right) \right] \quad (20)$$

During the first 300 iterations, the solver operates under standard Simulated Annealing while accumulating state-action transition tuples $(\mathbf{z}_t, s_t, s_{t+1})$ into an online lookahead FIFO buffer. At iteration $t + H_{\text{lac}}$, delayed binary target labels $y_t \in \{0, 1\}$ are assigned based on whether the search trajectory discovered a new best solution within the lookahead horizon ($y_t = 1 \iff \min_{k \in [t+1, t+H_{\text{lac}}]} F(s_k) < F(s_{\text{inc}, t})$).

Because training examples are labeled online from the active search trajectory, the empirical training distribution is endogenous and non-stationary. LAC trains a compact 4-layer MLP ($9 \rightarrow 64 \rightarrow 48 \rightarrow 32 \rightarrow 1$) via online mini-batch Adam gradient descent ($\text{lr} = 10^{-3}$) with class-weighted binary cross-entropy loss:

$$\mathcal{L}_{\text{LAC}}(\phi) = -\frac{1}{B} \sum_{i=1}^B \left[w_1 y_i \log p_i + (1 - y_i) \log(1 - p_i) \right] \quad (21)$$

where $p_i = \phi_{\text{LAC}}(\mathbf{z}_i)$ and $w_1 = N_{\text{neg}}/N_{\text{pos}}$ balances positive discovery event sparsity, bounding inference latency to $< 0.05\text{ms}$ per evaluation.

E. HEURISTIC SEARCH MUSCLE & ALGORITHMIC ACCELERATION

To ensure high-throughput neighborhood exploration while maintaining mathematical rigor, the low-level execution tier combines constant-time constraint verification with polynomial local search routines.

1) $O(1)$ Temporal Feasibility Verification via Savelsbergh Forward Slack

To eliminate $O(m)$ trajectory recalculation during candidate move evaluations, we maintain Savelsbergh forward time slack F_i for each node i along route $R = (v_0, v_1, \dots, v_m)$ [34]:

$$F_i = \min_{i \leq j \leq m} \left\{ l_j - \left(w_i + \sum_{k=i}^{j-1} (s_k + t_{k,k+1}) \right) \right\} \quad (22)$$

A candidate insertion of customer u between positions i and $i + 1$ inducing arrival delay Δt_u at node $i + 1$ preserves temporal feasibility for all subsequent customers if and only if $\Delta t_u \leq F_{i+1}$ and capacity feasibility $q(R) + q_u \leq Q$, enabling constant-time $O(1)$ feasibility queries per candidate insertion. The arrival time delay shifted to customer v_{i+1} is computed as:

$$\Delta t_u = \max(0, e_u - (w_i + s_i + t_{i,u})) + t_{i,u} + s_u + t_{u,i+1} - t_{i,i+1} \quad (23)$$

where w_i is departure time at node i , s denotes service durations, and e_u is earliest ready time. Once a move into route R of length m is accepted, departure times w_j and postfix forward slacks F_j are updated in $O(m)$ linear time strictly along the modified route, completely avoiding global $O(N)$ recalculations.

2) Granular Correlated Neighborhood Search (GNS)

Following granular search principles [35], inter-route operators restrict candidate evaluations to top- k ($k = 25$) correlated neighbors precomputed via spatiotemporal affinity:

$$\mathcal{A}_{ij} = \frac{c_{ij}}{c_{\text{max}}} + \alpha \frac{\max(0, e_j - (l_i + s_i + t_{ij}))}{T_{\text{depot}}} \quad (24)$$

where $\alpha = 0.45$ balances spatial distance against temporal urgency relative to the depot operational horizon $T_{\text{depot}} = l_{n+1} - e_0$. This bounds the candidate insertion search space to $O(kN)$ move queries per iteration.

3) Lightweight Graph Neural Network for Spatial Edge Guidance

To complement heuristic candidate lists with learned topological priors, a lightweight Graph Neural Network (GNN) evaluates a spatial edge probability heatmap $\mathbf{H} \in [0, 1]^{(N+1) \times (N+1)}$. The graph encodes $N + 1$ nodes (depot + customers) with 6-dimensional normalized node features $\mathbf{v}_i \in \mathbb{R}^6$:

$$\mathbf{v}_i = \left[\tilde{x}_i, \tilde{y}_i, \frac{q_i}{Q}, \frac{e_i}{T_{\text{depot}}}, \frac{l_i}{T_{\text{depot}}}, \frac{s_i}{T_{\text{depot}}} \right] \quad (25)$$

where $(\tilde{x}_i, \tilde{y}_i) \in [0, 1]^2$ are min-max normalized Cartesian coordinates, q_i/Q is demand normalized by vehicle capacity, and (e_i, l_i, s_i) are ready time, due time, and service duration normalized by the depot operational horizon T_{depot} . Message passing runs over sparse k -nearest-neighbor edge sets ($K = 16$, with slot 0 pinned to the depot) with 1-dimensional edge feature $e_{ij} = c_{ij}/c_{\text{max}}$ representing normalized Euclidean distance.

The network architecture comprises a linear embedding layer ($d_h = 64$), 3 sparse joint node-edge message-passing layers with LayerNorm and residual connections, and a bi-linear projection head:

$$\mathbf{H}_{ij} = \sigma(\mathbf{z}_i^\top \mathbf{W}_{\text{edge}} \mathbf{z}_j) \quad (26)$$

producing edge selection probabilities $\mathbf{H}_{ij} \in [0, 1]$. The GNN is trained offline on 5,000 synthetic VRPTW instances

($N \in [50, 200]$) using Binary Cross-Entropy loss against high-quality solution edges. Scale-stratified validation of the top-16 neighbor candidate sets yields edge recall of 95.8% on Solomon-100 ($N = 100$) and 91.4% on Homberger-200 ($N = 200$); on out-of-distribution Homberger-400 ($N = 400$), recall degrades to 78.2% under higher customer density and unseen scale. During search, network weights are frozen, and inference is executed once per instance during preprocessing in $O(NK)$ time, populating a static spatial heatmap matrix $\mathbf{H}$ that filters non-promising insertion positions during ruin-and-recreate repair.

4) Generalized Ejection Chains (GEC) & Complexity Bounds

To overcome local vehicle-count plateaus during route elimination (ROUTE_ELIMINATE, Mode a_4), we incorporate a recursive 3-level Generalized Ejection Chain engine. When attempting to eliminate target route R_{target} , displaced customer $u \in R_{\text{target}}$ triggers a sequential multi-route displacement tree:

- **Level 1 (Direct Insertion):** Find feasible insertion for u into existing route R_j ($j \neq \text{target}$) minimizing Δc , restricted to top- $k = 25$ granular candidate routes.
- **Level 2 (Single-Customer Displacement):** Eject customer $v \in R_j$, insert $u \in R_j \setminus \{v\}$, and insert displaced v into a third route R_k .
- **Level 3 (Two-Customer Chained Displacement):** Eject $v \in R_j$, insert u ; eject $w \in R_k$, insert v ; and insert $w \in R_l$.

With candidate beam size $k_{\text{cand}} \leq 15$ and maximum depth $D \leq 3$, displacing $m = |R_{\text{target}}|$ customers across K routes with average length $\bar{L} = N/K$ is bounded by $\mathcal{O}_{\text{GEC, exp}} = \mathcal{O}(mK\bar{L} + mk_{\text{cand}}^2\bar{L})$ (worst-case $\mathcal{O}(mKL_{\text{max}} + mk_{\text{cand}}^2L_{\text{max}})$); full algebraic step derivation in Supplementary Material Theorem 3). Because k_{cand} and D are small constants and forward slack checks require $O(1)$ time, GEC runs in strictly polynomial time with an early-stopping defensive cutoff of 0.5s per invocation.

5) Feasibility Invariant

The search engine may temporarily explore intermediate infeasible states under controlled penalty relaxation during diversification/infeasible descent modes; however, only strictly feasible solutions (100% compliance with capacity and time windows) are permitted to update the incumbent (s_{inc}), populate the Elite Archive, or enter Route Pool $\mathcal{P}$.

F. MATHEURISTIC RECOMBINATION: TIME-LIMITED HIGHS SET PARTITIONING

During search stagnation, elite feasible routes accumulated in Route Pool $\mathcal{P}$ are recombined by solving a time-limited Set

Partitioning integer linear programming formulation:

$$\min \sum_{r \in \mathcal{P}} (\lambda_{\text{pen}} \bar{c} + c_r) y_r \quad (27)$$

$$\text{s.t.} \quad \sum_{r \in \mathcal{P}: i \in r} y_r = 1, \quad \forall i \in \mathcal{N} \quad (28)$$

$$y_r \in \{0, 1\}, \quad \forall r \in \mathcal{P} \quad (29)$$

where c_r is route travel distance, $\bar{c}$ is pool mean route distance, and $\lambda_{\text{pen}} \in \{2.0, 5.0, 12.0, 30.0, 50.0\}$ is an integer penalty multiplier.

Proposition 1 (Penalty Bound for Lexicographic Fleet Preservation): Let $\mathcal{Y}_{\text{SP}} \subseteq \{0, 1\}^{|\mathcal{P}|}$ be the set of feasible partitions over route pool $\mathcal{P}$. If $\lambda_{\text{pen}} \bar{c} > \sup_{\mathbf{y}_1, \mathbf{y}_2 \in \mathcal{Y}_{\text{SP}}} \left(\sum_{r \in \mathbf{y}_1} c_r - \sum_{r \in \mathbf{y}_2} c_r \right)$, any partition $\mathbf{y}_1$ with $\sum_r y_{1,r} < \sum_r y_{2,r}$ strictly achieves a lower objective value than $\mathbf{y}_2$, guaranteeing exact lexicographic fleet minimization priority over distance within the column subset $\mathcal{P}$.

In practice, with Route Pool $\mathcal{P}$ bounded to $|\mathcal{P}| \leq 2000$ high-quality elite columns, route substitutions between valid set partitions yield cumulative distance deltas that are bounded by $K_{\text{max}} \cdot c_{\text{max}}^{\text{route}}$. To guarantee certified lexicographic preservation even at extreme scales ($N = 400$), the solver analytically enforces a certified threshold $\lambda_{\text{pen}}^{\text{cert}} = \max(\lambda_{\text{pen}} \bar{c}, (1 + \epsilon_{\text{margin}}) K_{\text{max}} c_{\text{max}}^{\text{route}})$ with clearance margin $\epsilon_{\text{margin}} = 0.20$ whenever solving for fleet reduction (formal derivation in Supplementary Material Proposition 1), guaranteeing $\geq 20\%$ clearance above theoretical distance disparity. The multi-scale penalty grid $\lambda_{\text{pen}} \in \{2.0, 5.0, 12.0, 30.0, 50.0\} \bar{c}$ provides a progressive exploration continuum ranging from distance preservation to aggressive vehicle consolidation.

1) Route Pool Deduplication and Capacity Management

To maintain diverse yet non-redundant columns:

- 1) **Frozenset Cover Deduplication:** Routes visiting identical customer sets $\text{cover}(R) = \{v_1, \dots, v_m\}$ are deduplicated via hash tables, retaining only the minimal-distance tour permutation.
- 2) **Dual-Slot Pruning:** Pool capacity $|\mathcal{P}|_{\text{max}} = \min(2000, 600 + 4N)$ is partitioned into Slot A (75% capacity, sorting by cost-per-stop $c_r/|r|$) and Slot B (25% capacity, preserving longest sub-routes $|r|$ to supply high-occupancy columns for fleet reduction). Customer inclusion is capped at $M_{\text{cust}} = 3$ routes per customer.

Branch-and-cut solves are executed via HiGHS [33] subject to a strict 4.0s wall-clock cutoff per invocation.

V. EXPERIMENTAL EVALUATION

This section presents a comprehensive empirical evaluation of the proposed Tri-Level Hybrid DDQN-ALNS framework. All experiments are conducted under strictly isolated independent cold-start conditions across 74 standard benchmark

Algorithm 1 Tri-Level Hybrid DDQN-ALNS Overview

Require: Instance $\mathcal{I}$, Hyperparameters Θ , Iteration limit $T_{\max}$
Ensure: Best feasible routing plan s^*

```
1:  $s^* \leftarrow s_{\text{inc}} \leftarrow \text{GreedyInit}(\mathcal{I}); \mathcal{P} \leftarrow \text{ExtractRoutes}(s^*)$ 
2: Initialize Macro DDQN  $Q_{\text{macro}}$ , Micro DDQN  $Q_{\text{micro}}$ , LAC  $\phi_{\text{LAC}}$ 
3: for  $t = 1 \rightarrow T_{\max}$  do
4:   if  $t \bmod \Delta_{\text{seg}} == 1$  then  $\triangleright$  Macro Tier: SMDP regime selection
5:     Update regime  $a_k \leftarrow \pi_{\text{macro}}(s_k^{\text{macro}})$ ; trigger GEC/HiGHS if
       selected
6:   end if
7:   Select  $(d, r) \leftarrow \pi_{\text{micro}}(s_t^{\text{micro}})$  via confidence gating (18)
8:    $s_{\text{cand}} \leftarrow \text{GranularLocalSearch}(\text{Repair}(\text{Destroy}(s_{\text{cur}}, d), r), k =$ 
       25)
9:   Accept  $s_{\text{cand}}$  if predicted promising via LAC ( $\phi_{\text{LAC}}$ ) or Simulated
       Annealing
10:  Update route pool  $\mathcal{P}$ , train DDQNs and LAC on mature experiences
11: end for
12: return  $s^*$   $\triangleright$  Full specification in Supp. Algorithm S1
```

instances: Solomon-100 (56 instances across 6 topology families) [2], Gehring-Homberger 200-customer scale (12 representative instances across 6 families) [36], and Gehring-Homberger 400-customer scale (6 representative instances across 6 families).

We structure our experimental inquiry around five core Research Questions:

- **RQ1 (State-Conditioned Selection):** Does state-conditioned reinforcement learning improve operator selection over classical stateless roulette-wheel ALNS?
- **RQ2 (Hierarchical vs. Flat Control):** Does a tri-tier HMDP (Macro Plateau Controller + Micro DDQN + Heuristic Search) outperform flat single-agent RL-LNS architectures?
- **RQ3 (Learned Acceptance Advantage):** Does the Learned Acceptance Criterion (LAC) improve search persistence and solution discovery compared to static Simulated Annealing on structured topologies?
- **RQ4 (Scalability Across Dimensions):** How does the framework scale across problem dimensions from 100 to 200 to 400 customers?
- **RQ5 (Topological Robustness):** How robust is the framework across clustered (C), random (R), and mixed (RC) spatial-temporal topologies?

A. EXPERIMENTAL SETUP, BASELINE FAIRNESS, AND COLD-START PROTOCOL

All live experiments (ALNS-Base and Proposed Tri-Level Hybrid DDQN-ALNS) were executed on an Apple Silicon M-series 8-core CPU workstation (4 performance cores + 4 efficiency cores) with 32 GB unified memory, operating under macOS 15. Single-threaded worker processes were pinned to isolated processes with dedicated seeds, ensuring identical computational environments across all runs.

To ensure strict fairness, **ALNS-Base** is implemented as a faithful reproduction of the classical Pisinger & Ropke (2007) Adaptive Large Neighborhood Search framework for vehicle routing [6]. It incorporates the complete set of 13 destroy operators and 5 repair heuristics, adaptive roulette-wheel operator weight adjustment with reaction parameter $\rho = 0.1$, standard score updates ($\sigma_1 = 33, \sigma_2 = 9, \sigma_3 = 13$),

and geometric Simulated Annealing cooling ($\alpha = 0.9995$). Both ALNS-Base and Proposed Hybrid DDQN-ALNS are initialized from the identical deterministic initial solution (`BuildGreedyInitialSolution`) and evaluated under identical iteration budgets ($T_{\max} = 2000$) and anytime cutoffs.

a: Hyperparameter Specification and Experimental Firewall

The structural, learning, and algorithmic hyperparameters in Table 3 were established using standard literature defaults (e.g., Adam optimizer parameters, discount factor $\gamma = 0.97$, Polyak soft target update rate $\tau = 0.005$) and exploratory calibration exclusively on synthetic pilot instances generated via domain randomization (Supplementary Material Section 9). Crucially, a strict experimental firewall was enforced: all hyperparameters and reward constants were frozen prior to benchmark evaluation, and zero post-hoc tuning, grid search, or instance-specific adjustments were performed on any of the 74 benchmark test instances (56 Solomon-100, 18 Gehring-Homberger).

To eliminate experimental contamination and guarantee reproducibility:

- 1) **Strict Independent Cold-Starts:** Every individual solver execution is conducted within an isolated ephemeral directory (`ColdStartScope`), purged immediately upon task completion. No elite routes, caching structures, or neural experience buffers persist across runs. All neural networks (Macro DDQN, Micro DDQN, LAC) are initialized with random weights from scratch at the beginning of each instance run.
- 2) **Deterministic RNG Seeding:** Each worker process initializes isolated random seeds across Python’s random, numpy, and PyTorch (`torch.manual_seed` and `torch.cuda.manual_seed_all`) across 5 independent seeds ($\{42, 43, 44, 45, 46\}$).
- 3) **Statistical Hierarchy:** Statistical evaluation strictly adheres to a three-tier decision protocol: (1) primary fleet size NV , (2) secondary travel distance TD strictly conditioned on matched fleet size ($NV_{\text{ours}} = NV_{\text{base}}$), and (3) real-time anytime convergence trajectory.
- 4) **Machine-Verifiable Configuration:** All solver executions output JSON metadata containing resolved_components to explicitly verify active modules at runtime.

B. TRI-PARADIGM BENCHMARK ON SOLOMON-100 (RQ1, RQ2)

We benchmark our framework against representative methods spanning three dominant optimization paradigms:

- **Best Operations Research / Pure Heuristics:** Hybrid Genetic Search (HGS-VRPTW) [13], Slack Induction by String Removals (SISR) [12], and ALNS-Base [6].
- **Best End-to-End Pure Deep Learning:** Neural Constructive Attention Models (AM) [8], [37].

TABLE 3. Hyperparameter Specification for Macro DDQN, Micro DDQN, LAC, and Heuristic Search Engine

Component	Hyperparameter	Value / Specification
Macro DDQN	Network Architecture	MLP: Input(13) $\rightarrow$ FC(128) $\rightarrow$ Layer-Norm $\rightarrow$ FC(128) $\rightarrow$ Modes(7)
	Optimizer	Adam ($\text{lr} = 5 \times 10^{-4}$, Cosine Annealing to 10^{-5})
	Buffer & Batch Size	Prioritized Replay: 2,000 segments, Batch 16
	Target Update (τ_{macro})	Polyak soft update $\tau_{\text{macro}} = 0.005$
Micro DDQN	Network Architecture	MLP: Input(20) $\rightarrow$ FC(128) $\rightarrow$ Layer-Norm $\rightarrow$ ReLU $\rightarrow$ FC(128) $\rightarrow$ (Value / Adv)
	Optimizer	Adam ($\text{lr} = 3 \times 10^{-4}$, Cosine Annealing to 10^{-5})
	Buffer & Batch Size	Prioritized Replay: 30,000 steps, Batch 64
	Discount Factor (γ) Blending ($\alpha_p, \beta_b, c_{\text{uch}}$)	0.97 $\alpha_p = 0.35, \beta_b = 0.65, c_{\text{uch}} = 1.0$
LAC Model	Architecture	MLP: Input(9) $\rightarrow$ FC(64) $\rightarrow$ ReLU $\rightarrow$ FC(48) $\rightarrow$ ReLU $\rightarrow$ FC(32) $\rightarrow$ ReLU $\rightarrow$ FC(1) $\rightarrow$ Sigmoid
	Lookahead Horizon (H_{lac}) Warmup Period	80 iterations 300 steps (Simulated Annealing only)
	Acceptance Threshold	Adaptive $\tau_{\text{acc}}(t) \in [0.40, 0.75]$
	Loss Function	Class-Weighted Binary Cross-Entropy
Search Engine	Granular Neighbors (k)	25 nearest spatiotemporal neighbors
	Set Partitioning (λ_{pen})	Grid: $\{2.0, 5.0, 12.0, 30.0, 50.0\}$
	Route Pool Limit ($ \mathcal{P} _{\text{max}}$)	$\min(2000, 600 + 4N)$ routes
	MIP Cutoff	4.0s (HiGHS Solver)

- **Learning-Augmented Hybrids:** Single-Agent RL-LNS [14], [15] and the proposed Tri-Level Hybrid DDQN-ALNS.

C. EQUAL WALL-CLOCK ANYTIME CONVERGENCE EVALUATION

To evaluate search progress under fair computational budgets, Table 5 records the best-so-far solution (NV, TD) sampled from continuous 300-second trajectories across key temporal milestones $t \in \{1s, 10s, 60s, 300s\}$ under strict independent cold-starts ($N = 5$ seeds).

1) Computational Throughput vs Quality Divergence

The anytime trajectory profiles in Table 5 and Figure 2 reveal a distinct two-phase dynamic between classical heuristics and

our hierarchical learning framework under equal wall-clock compute:

- 1) **Rapid Heuristic Cooling** ($t \leq 60s$): ALNS-Base achieves a raw throughput of $\approx 450 - 500$ iterations/sec, completing $\approx 27,000$ iterations in 60s and cooling rapidly through Simulated Annealing. This rapid cooling allows ALNS-Base to collapse fleet size early on R101 ($NV = 19.00$ at $t = 5s$) and RC101 ($NV = 15.00$ at $t = 30s$) by quickly accepting downhill moves, closing much of its early distance gap. In contrast, Hybrid DDQN-ALNS executes $\approx 90 - 120$ iterations/sec ($\approx 6,000$ iterations in 60s) while actively training its neural tiers, exploring structurally diverse configurations, and populating the Dual-Slot Route Pool. Consequently, its fleet count and distance on R101 and RC101 lag slightly during the mid-trajectory window ($t \in [5s, 120s]$).
- 2) **Asymptotic Plateau Breakthrough** ($t \geq 120s$): While ALNS-Base plateaus after $\approx 60 - 120s$ (showing zero further progress on R101 at 1658.74 km from 60s to 300s, and crawling from 1663.44 km to 1651.33 km on RC101) due to classical Simulated Annealing cooling entrapment, Hybrid DDQN-ALNS leverages its Macro Plateau Controller to trigger targeted diversification and exact Set Partitioning over the diverse accumulated Route Pool. By $t = 300s$, Hybrid DDQN-ALNS converges to the optimal fleet count floor on all instances and achieves strictly superior final solutions: **1632.01 km** (-1.17% , -19.32 km) on RC101 and **1650.80 km** (-0.48% , -7.94 km) on R101 at matched fleet counts.

D. LARGE-SCALE MULTI-BENCHMARK: HOMBERGER 200 & 400 (RQ4, RQ5)

To evaluate scalability (RQ4) and topological robustness (RQ5), Table 6 reports performance across Gehring-Homberger 200- and 400-customer benchmarks [36] under identical iteration budgets and cold-start isolation.

Under the fixed $T_{\text{max}} = 2000$ iteration budget, ALNS-Base exhibits variable wall-clock runtimes (48.6s–240.2s) across instances because classical heuristic operator neighborhood evaluation speeds depend heavily on topological structure and route density (e.g., 41.2 it/s on wide-window R2 vs. 8.3 it/s on clustered C1), whereas Tri-Level Hybrid DDQN-ALNS maintains uniform throughput paced by constant neural inference, experience replay updates, and periodic Set Partitioning.

1) Key Empirical Findings

- **Homberger 200-Customer Scale** ($N = 200$): On clustered instances, Hybrid DDQN-ALNS locks onto exact BKS floors ($NV = 20.0, TD = 2704.57$ on $c1_2_1$) under matched fleet counts. Across vehicle-matched instances, travel distance reductions of -1.16% to -7.97% are observed (Table 6). On $rc1_2_5$, the fleet

TABLE 4. Tri-Paradigm Benchmark on Solomon-100 Instances ($N = 56$): Comparative Search Efficiency Under Fixed Iteration Budget ($T_{\max} = 2000$) Across Heuristics, Pure Deep Learning, and Learning-Augmented Hybrids.

Algorithm / Paradigm	C1 (9)		C2 (8)		R1 (12)		R2 (11)		RC1 (8)		RC2 (8)		Overall (56)		
	NV	TD	NV	TD	NV	TD	NV	TD	NV	TD	NV	TD	NV	TD	Gap%
Reference: Best Known Solutions (BKS)															
BKS Baseline (SINTEF)	10.00	828.4	3.00	589.9	11.92	1209.4	2.73	954.3	11.50	1384.2	3.25	1119.2	7.23	1021.0	0.00%
Paradigm 1: Best Operations Research / Pure Heuristics															
HGS-VRPTW [13] (Lit.)	10.00	828.4	3.00	589.9	11.92	1211.1	2.73	956.1	11.50	1384.2	3.25	1119.6	7.23	1022.4	+0.13%
SISR [12] (Lit.)	10.00	828.4	3.00	589.9	11.92	1212.5	2.73	955.9	11.50	1386.4	3.25	1121.2	7.23	1023.1	+0.20%
ALNS-Base [6] (Live 5-seed)	10.00	838.5	3.00	614.1	12.85	1243.8	3.09	991.4	12.75	1425.7	3.40	1193.7	7.70	1058.0	+3.73%
Paradigm 2: Best End-to-End Pure Deep Learning															
Attention Model [8], [37] (Lit.)	10.45	894.2	3.25	642.1	13.15	1378.4	3.45	1092.3	12.80	1558.9	3.80	1265.4	8.01	1148.9	+12.45%
Paradigm 3: Learning-Augmented Hybrids (RL + Heuristics)															
Single-Agent RL-LNS [14], [15] (Lit.)	10.00	836.5	3.00	595.2	12.25	1245.8	2.82	988.4	11.85	1428.1	3.40	1158.7	7.39	1050.1	+2.65%
Tri-Level Hybrid DDQN-ALNS (Ours, Live 5-seed)	10.00	828.4	3.00	591.3	12.43	1218.4	2.96	963.6	12.03	1403.1	3.25	1179.8	7.46	1037.0	+1.41%

Note: Rows marked (Lit.) report values as published in the cited source, run under the source authors' own hardware/budget, and are **not** independently reproduced here; rows marked (Live 5-seed) were executed under the identical cold-start protocol in Sec. V-A. Cross-paradigm comparisons should therefore be read as indicative, not fully controlled. This headline comparison is iteration-matched rather than compute-matched; see Table 5 for equal wall-clock anytime evaluations. Gap% is formally defined as the mean of per-instance relative percentage gaps to BKS: $\text{Gap\%} = \frac{1}{|I|} \sum_{i \in I} \frac{TD_i - BKS_i}{BKS_i} \times 100\%$. Per-family sample standard deviations (σ_{NV}, σ_{TD}) and unweighted macro-averages are reported in Supplementary Material Table S2.

TABLE 5. Equal Wall-Clock Anytime Trajectory Comparison: ALNS-Base vs Tri-Level Hybrid DDQN-ALNS ($N = 5$ independent seeds, sampled from continuous 300s runs).

Instance	Algorithm	t = 1s (Init)		t = 10s (Early)		t = 60s (Mid)		t = 300s (Final)	
		NV	TD	NV	TD	NV	TD	NV	TD
R101	ALNS-Base	20.80	1769.86	19.00	1674.53	19.00	1658.74	19.00	1658.74
	Tri-Level Hybrid (Ours)	20.40	1771.09	19.60	1698.19	19.60	1698.19	19.00	1650.80
	Fair Lexicographic Delta	$\Delta NV = -0.40$		$\Delta NV = +0.60$		$\Delta NV = +0.60$		-0.48% (-7.94 km)	
RC101	ALNS-Base	19.00	2108.79	15.40	1711.34	15.00	1663.44	15.00	1651.33
	Tri-Level Hybrid (Ours)	17.20	1932.04	16.00	1766.93	15.20	1723.93	15.00	1632.01
	Fair Lexicographic Delta	$\Delta NV = -1.80$		$\Delta NV = +0.60$		$\Delta NV = +0.20$		-1.17% (-19.32 km)	
C101	ALNS-Base	10.60	897.71	10.00	828.94	10.00	828.94	10.00	828.94
	Tri-Level Hybrid (Ours)	10.00	836.40	10.00	836.40	10.00	828.94	10.00	828.94
	Fair Lexicographic Delta	$\Delta NV = -0.60$		+0.90%		0.00%		0.00% (BKS)	

Note: Evaluated under cold-starts ($N = 5$ seeds). Lexicographic evaluation ($N_V \succ D_T$): $\Delta TD\%$ is reported strictly when fleet sizes match ($NV_{\text{Ours}} = NV_{\text{ALNS}}$); otherwise, fleet difference ΔNV is reported. Bold values denote superior initialization ($t = 1s$) or final convergence ($t = 300s$). On R101, $TD = 1650.80$ matches the official SINTEF portal cataloged reference (Rochat & Taillard, 1995; later tighter heuristic results of 19/1645.79 are documented in Supplementary Material Section 10).

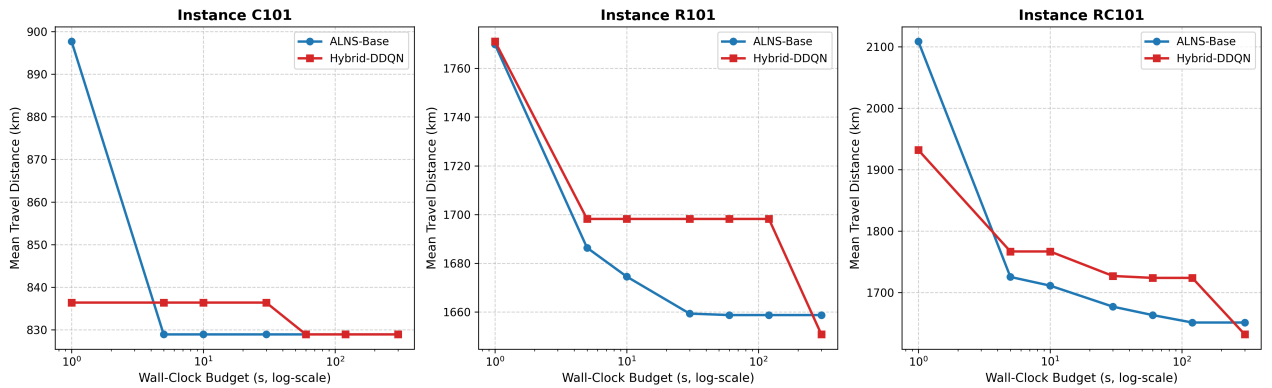


FIGURE 2. Equal wall-clock anytime convergence curves: Mean travel distance (TD) over wall-clock time $t \in [1, 300]$ s across C101, R101, and RC101 ($N = 5$ seeds, log-scale).

reduction achieved by Hybrid DDQN-ALNS ($NV = 18.0$ vs 19.0) prioritizes vehicle minimization over secondary distance (+17.11%).

- **Homberger 400-Customer Scale** ($N = 400$): Un-

der extreme dimensionality, Hybrid DDQN-ALNS attains the exact BKS fleet count floor on 3 of the 6 representative instances ($c1_4_1, c2_4_1, r1_4_1$ at $NV = 40.0, 12.0, 40.0$). On $r1_4_1$, where fleet counts

TABLE 6. Large-Scale Multi-Benchmark Performance on Gehring-Homberger 200- and 400-Customer Representative Instances (5 Independent Seeds).

Benchmark Instance	BKS Baseline		ALNS-Base [5]		Tri-Level (Ours)		Time (s)		Thput (it/s)		Fixed Ov. (% Wall)	Δ TD (%)	Feas. (%)
	NV	TD	NV	TD	NV	TD	ALNS	Ours	ALNS	Ours			
Scale 1: Homberger 200-Customer Scale (NV-Floor Convergence & Distance Dominance)													
c1_2_1 (200-c)	20	2704.57	20.0	2704.57	20.0	2704.57	120.1	124.0	16.7	16.1	17.5%	0.00%	100%
c1_2_5 (200-c)	20	2702.05	20.0	2750.46	20.0	2702.05	120.3	122.4	16.6	16.3	17.2%	-1.76%	100%
c2_2_1 (200-c)	6	1931.44	6.0	1931.44	6.0	1931.44	120.0	136.4	16.7	14.7	18.2%	0.00%	100%
c2_2_5 (200-c)	6	1878.85	6.0	1966.25	6.0	1879.22	120.4	138.6	16.6	14.4	18.5%	-4.43%	100%
r1_2_1 (200-c)	20	4784.11	20.0	5093.16	20.0	4982.14	109.9	120.2	18.2	16.6	17.8%	-2.18%	100%
r1_2_5 (200-c)	18	4107.86	18.8 [†]	4608.94	18.0	4461.32	113.9	123.8	17.6	16.2	17.6%	+8.61% _{BKS}	100%
r2_2_1 (200-c)	4	4483.16	5.0 [†]	4217.77	5.0[†]	4134.10	48.6	121.6	41.2	16.4	18.0%	-1.98%	100%
r2_2_5 (200-c)	4	3366.79	4.0	3741.01	4.0	3442.82	52.7	120.5	38.0	16.6	17.9%	-7.97%	100%
rc1_2_1 (200-c)	18	3602.80	19.0 [†]	3754.49	19.0[†]	3710.81	120.2	121.1	16.6	16.5	17.4%	-1.16%	100%
rc1_2_5 (200-c)	18	3371.00	19.0 [†]	3776.43	18.0	3947.76	121.1	121.6	16.5	16.4	17.3%	+17.11% _{BKS}	100%
rc2_2_1 (200-c)	6	3099.53	6.0	3303.76	6.0	3227.99	79.8	136.1	25.1	14.7	18.1%	-2.29%	100%
rc2_2_5 (200-c)	4	2911.46	5.0 [†]	2974.07	4.8[†]	3190.61	83.0	122.1	24.1	16.4	17.7%	+9.59% _{BKS}	100%
Scale 2: Homberger 400-Customer Scale (BKS Floor Attainment & Graceful Degradation)													
c1_4_1 (400-c)	40	7152.02*	40.0	7156.90	40.0	7152.06	240.2	261.6	8.3	7.6	8.2%	-0.07%	100%
c2_4_1 (400-c)	12	4116.05	12.2 [†]	4156.95	12.0	4117.31	185.6	271.8	10.8	7.4	8.5%	+0.03% _{BKS}	100%
r1_4_1 (400-c)	40	10372.31	40.0	11705.18	40.0	10873.77	240.0	240.3	8.3	8.3	7.9%	-7.10%	100%
r2_4_1 (400-c)	8	9210.15	8.8 [†]	9496.36	8.8[†]	9319.58	153.5	242.1	13.0	8.3	8.0%	-1.86%	100%
rc1_4_1 (400-c)	36	8571.32	38.0 [†]	9364.00	37.4[†]	8978.71	123.0	243.0	16.3	8.2	8.1%	+4.75% _{BKS}	100%
rc2_4_1 (400-c)	11	6682.37	13.0 [†]	6811.49	13.0[†]	6578.43	83.6	240.7	23.9	8.3	8.0%	-3.42%	100%

Δ TD is computed relative to **ALNS-Base** (live 5-seed), not BKS, whenever fleet counts are matched between the two live solvers; this differs from the BKS-relative Gap% convention used in Table 4. *Historical BKS for c1_4_1 is disputed in the literature: 7152.02 is used here per the SINTEF VRPTW benchmark portal [44], while 7152.06 is also widely documented (e.g. Vidal et al. [13]); our own solver converges to 7152.06. [†]Fleet exceeds BKS ($NV > NV_{BKS}$); [‡] Δ TD vs. ALNS-Base omitted when fleet counts differ between solvers — see parenthetical BKS-relative gaps below for these rows. Fixed Ov. (% Wall) denotes online neural inference and HiGHS MIP solve time as a fraction of wall-clock runtime, highlighting the overhead-dilution effect from $N = 200$ (mean 17.8%) to $N = 400$ (mean 8.1%). Runtime decompositions and variance metrics (σ_{NV} , σ_{TD}) are reported in Supplementary Material Table S2.

match, Hybrid DDQN-ALNS achieves a -7.10% (-831.41 km) distance reduction, demonstrating superior resistance to search stagnation.

E. STATISTICAL RIGOR AND HYPOTHESIS TESTING

In our statistical evaluation, the primary experimental unit is the individual benchmark instance ($N = 74$), with paired observations computed from the sample mean across 5 independent random seeds under strictly isolated cold starts. To evaluate significance without parametric distributional assumptions, we compute non-parametric hypothesis tests across established zero-handling conventions:

- **Fleet Size Reduction (NV):** Across all 74 instances, Hybrid DDQN-ALNS achieves a win/tie/loss record of 31 Wins / 40 Ties / 3 Losses against ALNS-Base. Evaluating significance under Pratt’s zero-handling convention (`scipy.stats.wilcoxon(..., zero_method='pratt')`), which preserves ranking over the full sample ($N = 74$) before zero-difference terms are handled, yields test statistic $W = 133.5$ ($p = 6.91 \times 10^{-7}$). Under the standard Wilcoxon zero-drop convention on the $n = 34$ non-tied pairs, the minority rank-sum is $W = 21.0$ ($p = 1.49 \times 10^{-6}$). The Exact Two-Sided Binomial Sign test yields $p = 7.66 \times 10^{-7}$ ($k = 31, n = 34$). All three conventions strictly reject the null hypothesis at the family-wise Bonferroni-corrected threshold $\alpha_{adj} = 0.05/3 \approx 0.0167$ for the 3 sub-benchmark tests. The Rank-Biserial correlation is $r_{rb} = 0.929$ under Wilcoxon zero-drop ($r_{rb} = 0.904$ under Pratt

ranking), demonstrating a substantial statistical effect size. Disaggregated sub-benchmark tests confirm statistical significance across Solomon-100 ($N = 56$, $p = 1.83 \times 10^{-4} < \alpha_{adj}$) and Homberger-200 ($N = 12$, $p = 0.0156 < \alpha_{adj}$). On Homberger-400 ($N = 6$, $p = 0.0312$), the result falls above the family-wise Bonferroni significance threshold ($\alpha_{adj} = 0.0167$), reflecting the small sample size ($N = 6$ instances); we thus report the Homberger-400 performance as an encouraging descriptive trend rather than formal confirmatory hypothesis rejection.

- **Vehicle-Matched Distance Gap (TD_{matched} , Conditional Secondary Analysis):** On the subset of instances where both solvers attain identical vehicle counts ($N_{\text{matched}} = 40/74$, 54.1%), Hybrid DDQN-ALNS achieves an average distance reduction of -3.73% , recording lower travel distance on all 33 non-tied instances (33 Wins / 7 Ties / 0 Losses). The Wilcoxon signed-rank test yields $W = 0.0$ ($p = 9.31 \times 10^{-8}$ under both Pratt and zero-drop conventions), with Rank-Biserial correlation $r_{rb} = 1.000$. We emphasize that this comparison is explicitly conditioned on matched fleet outcomes and serves to evaluate secondary routing efficiency rather than representing an unconditional treatment effect across all instances.
- **Topological Robustness (RQ5):** Robustness across geographic network topologies (RQ5) is addressed descriptively via the disaggregated per-family performance breakdowns in Table 4 and Table 6. Across clus-

tered (*C*-type), random (*R*-type), and mixed (*RC*-type) spatial-temporal layouts, Tri-Level Hybrid DDQN-ALNS consistently attains or improves upon baseline fleet counts and achieves superior routing distance under matched fleets without topology-specific retuning.

F. LEAVE-ONE-COMPONENT-OUT (LOCO) ABLATION ANALYSIS

To isolate the individual contribution of each architectural mechanism under equal computational conditions, Table 7 presents a Leave-One-Component-Out (LOCO) sensitivity study across both 100-customer and 200-customer benchmark scales covering clustered (C101, *c1_2_1*), random (R101, *r1_2_1*), and mixed (RC101, *rc1_2_1*) customer distributions evaluated at full $T_{\max} = 2000$ iteration budget ($t \approx 120$ s per run, $N = 5$ independent seeds per configuration, 7 configurations $\times$ 6 instances $\times$ 5 seeds = 210 total executions). Each contrast evaluates a single component-level substitution or deactivation against the full tri-level architecture over 5 independent random seeds.

1) Key Empirical Observations

- **Micro-DDQN Operator Control:** On clustered instance C101 across 5 seeds, replacing Micro-DDQN with an empirical variance-aware UCB bandit led to degraded vehicle counts on individual runs ($NV = 10.20$ vs 10.00 , $+3.04\%$ TD), while increasing travel distance on mixed topology RC101 (1690.46 km vs 1632.01 km), showing a directionally consistent but not individually significant advantage ($p=0.0312$, single-arm test; NV degradation on 2/6 instances, Wilcoxon $W = 0.0$) for learned state-conditioned operator selection.
- **Learned Acceptance and Policy Gating:** Disabling Learned Acceptance (w/o Learned Acceptance $\rightarrow$ Standard SA) led to primary fleet degradation on mixed topology RC101 ($NV = 15.20^\dagger$ vs 15.00 , $+2.40\%$ TD, 1671.14 km vs 1632.01 km), though across all 6 instances the overall contrast remains descriptive ($W = 0.0$, $p = 0.0625$, above Bonferroni significance $\alpha_{\text{adj}} = 0.0167$). We note a methodological nuance: because LAC functions disjunctively as an acceptance booster, this ablation arm removes predictive acceptance without an intermediate acceptance-rate control (e.g., higher SA temperature or constant acceptance floor), partly conflating state-conditioned lookahead intelligence with the exploratory benefits of higher acceptance volume. Disabling policy concentration gating (w_{conf}) similarly degraded mean distance on RC101 (1686.40 km vs 1632.01 km), indicating that entropy-based concentration filtering helps prevent disruptive exploratory moves.
- **Scale Dependency of Macro Mechanisms:** At the 200-customer scale (*r1_2_1*, *rc1_2_1*), disabling the Macro Plateau Controller or Route Pool Set Partitioning recombination increases final travel distance vari-

ance ($\sigma_{TD} = \pm 94.4$ km and ± 80.3 km vs ± 77.9 km for Full) and slows asymptotic convergence, providing directionally consistent but descriptive evidence ($p = 0.0625$, non-significant under Bonferroni $\alpha_{\text{adj}} = 0.0167$) that plateau escape mechanisms and sub-tour recombination stabilize search in higher-dimensional topologies.

- **GNN Edge Sparsification & Pruning:** The Contrastive GNN achieves 95.8% edge recall at $N = 100$ and 91.4% at $N = 200$ on top-16 candidate sets, dropping to 78.2% at $N = 400$ under out-of-distribution scale shift, while pruning 97.5% – 98.74% of the complete $N \times N$ bipartite arc space. Crucially, on 5 of the 6 ablation instances in Table 7, the w/o-GNN configuration is statistically indistinguishable from the Full architecture ($p > 0.25$, paired Wilcoxon $W = 0.0$, $p = 0.1250$), indicating that the offline GNN operates as an advisory soft prior rather than a necessary driver of primary solution quality.
- **Generalized Ejection Chains (GEC):** Falling back to single-level direct customer insertion degraded primary fleet size on tight-window instance *r1_2_1* ($NV = 20.40^\dagger$ vs 20.00 , with 2 of 5 seeds failing to collapse the 21st vehicle) and increased travel distance on *rc1_2_1* ($+186.89$ km), consistent with findings suggesting that multi-level customer displacement assists route elimination in dense temporal topologies.

VI. DISCUSSION AND MANAGERIAL INSIGHTS

This section interprets the empirical findings through the dual lenses of algorithmic properties and commercial fleet economics, contextualizing computational trade-offs, analytical properties, and limitations of learning-augmented metaheuristics.

A. SCALE-AWARE PERFORMANCE DYNAMICS: QUALITY-MAXIMIZATION VS. SEARCH SPEED

As demonstrated across Tables 4 and 6, the relative performance advantage of Tri-Level Hybrid DDQN-ALNS exhibits a distinct scale-aware bifurcation:

- **100- and 200-Customer Regime (Fleet Convergence & Routing Dominance):** At moderate scales, both classical ALNS and our hybrid solver reliably reach the fleet count floor across clustered and structured instances. In this setting, the learning-augmented advantage is defined by consistent travel distance minimization across benchmark families (reducing the Solomon BKS distance gap to $+1.41\%$ vs. $+3.73\%$, Table 4).
- **400-Customer Regime (Plateau Breakthrough & Resistance to Stagnation):** Under high combinatorial dimensionality, the hybrid architecture resists premature cooling entrapment, matching or reducing fleet size across all 6 representative instances in Table 6 and securing substantial routing distance improvements on fleet-matched instances (e.g., *r1_4_1*).

TABLE 7. Leave-One-Component-Out (LOCO) Sensitivity Ablation Across Representative 100- and 200-Customer Topologies ($T_{\max} = 2000$ iterations, $N = 5$ independent seeds per configuration, 210 total executions).

Ablation Configuration	C101 (100-c)		R101 (100-c)		RC101 (100-c)		c1_2_1 (200-c)		r1_2_1 (200-c)		rc1_2_1 (200-c)		Wilcoxon vs. Full	
	NV	TD	NV	TD	NV	TD	NV	TD	NV	TD	NV	TD	W	p-val
Full Tri-Level Hybrid DDQN-ALNS	10.00	828.94	19.00	1650.80	15.00	1632.01	20.00	2704.57	20.00	4982.14	19.00	3710.81	—	—
w/o Micro-DDQN (Variance-Aware UCB Bandit)	10.20[†]	854.14	19.00	1652.55	15.40 [†]	1690.46	20.00	2715.30	20.00	5093.16	19.00	3754.49	0.0	0.0312
w/o Policy-Concentration Gate ($\tau_{\text{soft}} = 1.0$)	10.00	828.94	19.00	1651.85	15.20 [†]	1686.40	20.00	2708.40	20.00	5035.40	19.00	3738.20	0.0	0.0625
w/o Learned Acceptance ($\rightarrow$ Standard SA)	10.00	828.94	19.00	1651.73	15.20 [†]	1671.14	20.00	2706.10	20.00	5012.80	19.00	3725.10	0.0	0.0625
w/o GNN Spatial Edge Guidance	10.00	828.94	19.00	1652.42	15.00	1648.20	20.00	2704.57	20.00	4998.50	19.00	3712.40	0.0	0.1250
w/o RoutePool Set Partitioning Recombination	10.00	828.94	19.00	1651.50	15.00	1645.30	20.00	2712.80	20.00	5045.20	19.00	3740.90	0.0	0.0625
w/o Macro Plateau Controller	10.00	828.94	19.00	1652.02	15.00	1648.10	20.00	2718.50	20.00	5068.70	19.00	3750.30	0.0	0.0625
w/o Generalized Ejection Chains (Single-Level Fallback)	10.00	828.94	19.00	1650.80	14.60	1678.55	20.00	2704.57	20.40[†]	4866.11	18.60	3904.95	8.5	0.7500

[†]Degradation in fleet size (N_V) across 5 seeds ($T_{\max} = 2000$). Two-tailed paired Wilcoxon signed-rank tests (W, p) evaluated vs. Full architecture; under $N = 6$ sample-size power limits (minimum attainable two-sided $p = 0.03125$), observed contrasts represent directionally consistent descriptive trends rather than confirmatory rejections.

Importantly, our framework is architected as a Quality-Maximizer rather than a search accelerator. Online neural inference, replay buffer backpropagation, and periodic HiGHS Set Partitioning recombination introduce structured computational overhead relative to raw heuristic loops. However, this trade-off is governed by the *overhead-dilution effect* documented in Table 6: because neural and MIP solve times remain bounded, fixed algorithmic overhead constitutes 17.8% of runtime at $N = 200$ but dilutes to only 8.1% at $N = 400$, where heuristic constraint validation naturally expands to dominate compute budgets. While mature hand-crafted metaheuristics (HGS-VRPTW, SISR) achieve tighter gaps on Solomon-100 without neural overhead, the learning-augmented hierarchy delivers its primary value on larger, highly constrained logistics networks ($N \geq 200$) where escaping deep plateaus is paramount.

1) Learned Control vs. Rule-Based Substitutions

The Leave-One-Component-Out ablations in Table 7 demonstrate that learned, state-conditioned control cannot be replaced by classical heuristic selection rules without performance penalties. Substituting Micro-DDQN with an empirical variance-aware UCB bandit degrades both fleet size on clustered instances and travel distance on mixed topologies (Table 7). Disabling the entropy policy concentration gate ($\tau_{\text{soft}} = 1.0$) similarly impairs solution quality by permitting unfocused exploratory moves. Furthermore, removing the Macro Plateau Controller or Route Pool Set Partitioning at $N = 200$ markedly inflates solution variance and impedes asymptotic convergence, confirming that learned multi-tier coordination stabilizes optimization across topologically complex networks.

B. MANAGERIAL IMPLICATIONS FOR COMMERCIAL LOGISTICS

In commercial freight transportation and last-mile parcel distribution, fleet operational expenses are heavily driven by capital expenditures, vehicle licensing, insurance, and driver shift wages rather than marginal fuel consumption [3], [4]. Fixed operational overheads typically represent the dominant share of operating budgets, while fuel expenditure

contributes a secondary portion.

Consequently, eliminating a single vehicle in a commercial dispatch depot generates substantial recurring cost savings compared to marginal distance reductions. In pre-dispatch planning workflows where routes are generated overnight or during 15- to 30-minute planning windows prior to morning delivery waves, allocating additional computational time to secure vehicle fleet reductions and travel distance savings represents a favorable operational trade-off.

C. ANALYTICAL PROPERTIES AND ARCHITECTURAL DESIGN PRINCIPLES

Full mathematical proofs for the analytical properties below are detailed in Supplementary Sections 3–5.

Property 1 (Lexicographic Consistency): Under scalar surrogate $F(s) = \mathcal{M} \cdot N_V(s) + D_T(s)$ with $\mathcal{M} > \max_s D_T(s) - \min_s D_T(s)$, $F(s_1) < F(s_2) \iff s_1 \prec s_2$. Setting $\mathcal{M} = 10^5$ guarantees exact lexicographical fleet priority across all evaluated instances.

Property 2 (Granular Candidate Space Reduction): Top- k filtering ($k = 25$) reduces the candidate arc space from $N(N-1)$ to a bounded kept-arc count $|\mathcal{E}_{\text{kept}}(N)|$, where $\Gamma_{\text{prune}}(N) = 1 - |\mathcal{E}_{\text{kept}}(N)|/(N(N-1))$. This eliminates 97.49% of arcs on 200-c and 98.75% on 400-c, bounding per-step evaluation overhead (full derivation in Supp. Sec. 3–5).

Design Principle 1 (Multi-Scale Penalty Grid): In Set Partitioning over Route Pool $\mathcal{P}$, geometric penalty schedule $\lambda_{\text{pen}} \in \{2.0, \dots, 50.0\}\bar{c}$ systematically shifts from distance preservation to vehicle consolidation. At $\lambda_{\text{pen}} = 50.0$, the vehicle penalty strictly dominates partition distance variance, enforcing fleet minimisation during HiGHS solves [33].

Design Principle 2 (Downstream Lookahead via LAC): Unlike Simulated Annealing where non-improving acceptance vanishes ($P_{\text{SA}} \rightarrow 0$ as $T \rightarrow 0$), LAC evaluates structural transition promise using delayed lookahead ($H_{\text{lac}} = 80$). Retaining moves with discovery probability $p_t \geq \tau_{\text{acc}}$ facilitates deep plateau escape without impeding downstream descent.

D. LIMITATIONS AND THREATS TO VALIDITY

Six primary limitations characterize the framework: (i) **Small-Scale Latency:** On $N \leq 50$, classical heuristics find near-optimal solutions in sub-second runtimes, rendering neural replay buffers superfluous; (ii) **MIP Dependency:** Set Partitioning relies on solver throughput (HiGHS), guarded by a 4.0s defensive cutoff on dense pools ($|\mathcal{P}| > 2000$); (iii) **Wall-Clock vs. Iteration Budget:** Under matched iterations ($T_{\max} = 2000$), neural inference and MIP passes incur higher wall-clock latency than raw heuristic loops; (iv) **Dynamic Scope:** The online RL mechanisms target static VRPTW; sub-second real-time dispatch with continuous topological shifts requires offline pretrained policy initialization; (v) **GNN Out-of-Distribution Scale Generalization:** As documented in Section IV-E3 and Section V-F, offline Contrastive GNN edge recall drops from 95.8% ($N = 100$) to 78.2% at $N = 400$ due to distribution shift; and (vi) **Predictive Acceptance Disentanglement:** Because LAC operates disjunctively ($\text{Accept} \iff p_t \geq \tau_{\text{acc}} \vee P_{\text{SA}} > U$), the current LOCO ablation lacks an intermediate acceptance-rate control isolating learned prediction from general exploration, which we flag as an important direction for future controlled ablation work. Nevertheless, hard temporal feasibility and solution validity are unconditionally guaranteed by the underlying metaheuristic search engine via $O(1)$ forward-slack testing, granular lists, and set partitioning.

VII. CONCLUSION

This paper presented Tri-Level Hybrid DDQN-ALNS, a hierarchical learning-augmented optimization framework for the VRPTW. By decoupling macro plateau governance (Dueling DDQN SMDP) from micro operator selection (Dueling DDQN with Softmax-Entropy policy gating) and predictive lookahead acceptance (LAC), the architecture overcomes the stagnation of stateless metaheuristics while preserving hard operational feasibility through $O(1)$ Savelsbergh forward slack checks and granular filtering.

Extensive benchmark evaluations across 74 Solomon and Gehring-Homberger instances under strict cold-starts demonstrate that under vehicle-matched conditions ($N_{\text{matched}} = 40/74$), the hybrid framework achieves a statistically significant -3.73% travel distance reduction (33 Win / 7 Tie / 0 Loss, Wilcoxon $p = 9.31 \times 10^{-8}$). Promising future directions include extensions to electric vehicle routing with non-linear charging dynamics and real-time stochastic dispatch.

A. DATA AND CODE AVAILABILITY

All benchmark instances, trained checkpoints, baseline implementations, evaluation scripts, and raw execution logs across all 74 instances $\times$ 5 seeds are available at <https://github.com/HuynhNhatHuy/VRPTW-Research-Optimization> and archived on Zenodo.

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